Zoology Legends

Bio203

Current Final Term Papers 2023

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Zoology Legends

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Mcqs:

(Mostly from Files)

- PCR _____ **Polymerase chain reaction**
- RFLP_____ **Restriction fragment length polymorphism**
- Random Amplified Polymorphic DNA_____**RAPD**
- Amplified fragment length polymorphism_____**AFLP**
- which is type of microscope having beams of electrons_____ **Electron Microscope**
- Which type of pcr is done inside the cell_____**In Situ PCR**
- The type of pcr which defines bacterial yeast clones is..... **Colony PCR**
- STR stands for.....**Short Tandem Repeats**
- How many Screening Methods SNPs have.....**2**

Zoology Legends

- WGA PCR stands for..... **Whole Genome Amplification Polymerase Chain Reaction**
- **To amplify one strand of the original DNA more than the other, a special type of PCR is used which is known as _____.Asymmetric Pcr**
- Type of pcr which assembel many pieces into one..... **Assembly Pcr**
- Immunosorbent Electron Microscopy contains..... antibodies. **Both (monoclonal and polyclonal)**
- In Gel Electrophoresis molecules movement depend on ..... **Size**
- 10% DMSO Works well as a.....**Freezing Medium**
- Antibodies recognize a unique molecule called..... **Antigen**
- How many % attribute make a men unique. **0.1%**
- Multiplex-PCR consists of multiple primer sets within a ------**Single**------ PCR

mixture to produce amplicon of varying sizes that are specific to different DNA sequences.

- The PCR is preceded by a reaction using reverse transcriptase to convert

  ---------- **RNA to cDNA**

- Real Time PCR is a technique in which --------- **fluorophores** --------- bind to specific target regions of amplicons to produce fluorescence during PCR.
- Allele-specific PCR used for identifying of ---------- **SNPs**.
- **Colony PCR**_ The screening of bacterial or yeast clones for correct ligation or plasmid products.
- **IN SITU PCR** is a polymerase chain reaction that actually takes place inside the cell on a slide

- DNA profiling or DNA fingerprinting is a -------- **forensic technique** used to identify individuals by characteristics of their DNA
- PCR has also facilitated research in detection of ----------**Pathogens**--------- in plants, animals and environment
- **Type 111** _ systems are of little value for gene manipulation
- A suitable system was proposed by ---------- **Smith and Nathans (1973)**
- Type II endonucleases recognize and cleave DNA within particular sequences of **four to eight** nucleotides which have a twofold axis of rotational symmetry i.e. referred as ---------- **palindromes**

- Six base recognition site occurs every ---------- **46 (4096) bp**
- **DNA ligase** is extracted from phage T4 infected E. coli
- The enzyme responsible for removing phosphate groups from many types of molecules including DNA____ **Alkaline Phosphatase**
- The Southern blot is used to detect the presence of ---------- in a sample by a molecular probe__ **particular piece of DNA**
- Northern blot is used for Study of **gene expression** in eukaryotic cells
- Diseases which are included in prenatal ……. **Inherited disorders**

- In Paleontology ----- **Molecular** --- is studied by PCR.
- Nested PCR primers_ **Two pairs (instead of one pair) of PCR primer**
- Protein of interest is detected and localized using a specific..... **Antibody**
- wet transfer is recommended for large protein.....**100 kd**
- RAPD stands for ....**Random amplified polymorphism DNA**
- Southern blotting fragments is for.... **DNA**
- How many different SNP screening..... **2**
- Enzyme responsible for joining DNA fragments.....**DNA Ligase**
- cohesive fragment of DNA produced **EcoR1**
- Suicide PCR is typically used in .....**paleo genetics**

- How many% attribute men unique. **0.1%**
- ..... Molecules movement depend on size. **Gel electrophoresis**
- Immunosorbent electron contain..... **Both (monoclonal and polyclonal)**
- Which polymerase enzyme is used in pcr___(**Taq polymerase**)
- Initial temperature of denatured step of pcr_____**(93 to 98)**
- Which pcr take place inside the cell **(in situ pcr)**
- Which DNA probe is used in QRT_pcr **(taqman)**
- In Mimi primer pcr which polymerase is used **(S_Tbr)**
- An enzyme which is responsible to remove phosphate group from many molecules including DNA___**(Alkaline phosphate)**

- Non-radioactive probe of southern blotting **(biotin)**
- Which method is used for detection of RNA__**(northern blotting)**
- People can be grouped on the basis of **(SNP profile)**
- Pcr is used in__(gene cloning, gene screening,forensic, **All given options**)
- RAPD stands for__**(random amplified polymorphic DNA)**
- Which technique is used to differentiate between Mobile phase and stationary phase **(chromatography)**
- WGA stands for__**(whole genome amplification pcr)**
- Primer length_____**(18 to 30 base pair)**

Short & Long (Repeated) Questions:

(Most Answers are Taken from PPTs)

dNTPs

Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification.

dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 μM.

Nested and Multiplex PCR

NESTED PCR	MULTIPLEX PCR
Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR.	Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using

Very low probability of nonspecific amplification.	more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences

In situ PCR

- In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide.
- In situ PCR amplification can be performed on fixed tissue or cells.
- Applies the methodology of hybridization of the nucleic acids.
- Allows identification of cellular markers
- Limited to detection of non-genomic material such as RNA.

Type 1 of Restriction and Modification (R-M)

Type I

♣ Type I systems were the first to be characterized from E. coli K 12

♣ The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit

♣ Type I systems are of little value for gene manipulation

Negative control

 Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

 The PCR product should be of the expected size.

Nomenclature of The Smith and Nathan's (1973)

♣ A suitable system was proposed by Smith and Nathans (1973)

♣ The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet

♣ E. coli = Eco

♣ H. influenzae = Hin

♣ Strain identification is written as Eco K

♣ In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae

♣ HindI , HindII

dNTPs or Assembly PCR:

Deoxynucleoside triphosphates (dNTPs)	ASSEMBLY PCR
Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification. dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP,	Assembly PCR used to assemble two or more pieces of DNA into one piece. Assembly PCR **PRINCIPLE** Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more

dGTP, dTTP) in the PCR is usually 200 µM.	pieces of DNA into one piece.

Application of PCR in microbiology.

- Identification of bacteria:
- Slow growing bacteria
- Bacterial antibiotic resistance genes.
- VIRUS: almost all viruses identification and viral load.
- FUNGUS
- Aspergillus spp
- Candida spp
- Plasmodium spp
- Trypanosoma spp

Steps of PCR

THREE STEPS

- Denaturation
- Annealing
- Extension S

Nested primers.

Nested primers (as they lie / are nested within the first fragment).

Asymmetric PCR

- Asymmetric PCR is used to amplify one strand of the original DNA more than the other.
- It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.
- PCR is carried out as usual, but with a great excess of one primers for the chosen strand.

 Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required.

Advantages of PCR

 Specific

 Simple, rapid, relatively inexpensive

 Amplifies from low quantities

 Works on damaged DNA

 Sensitive

 Flexible

Types of Restriction and modification (R-M) system

Types of restriction and modification (R-M) system

♣ At least **four** R-M systems are known

I. Type I
II. Type II
III. Type III
IV. Type IIs

3 Steps of designing Primers (Ans from Google)

How to Design Primers for PCR and quantitative real time PCR (qPCR)

- ♣ Keep the melting temperatures (T_m) of each primer pair within 2°C of one another
- ♣ Use an annealing temperature (T_a) 3-5°C below the melting temperature
- ♣ Keep primers between 18-22 base pairs long
- ♣ Design primers with a GC content of 35-65%.

Ligation Mediated PCR

 Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA.

 PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

 DNA sequencing

 Genome walking

 DNA foot printing

Methodology of Assembly PCR

 It involves an initial PCR with primers that have an overlap and a second PCR using the products as the template that generates the final full-length product.

 Assembly PCR used to assemble two or more pieces of DNA into one piece.

What reagents are used in standard PCR? (3)

Generally, a complete polymerase chain reaction (PCR) requires five basic reagents, including

- DNA template,
- Forward and reverse primers,
- DNA polymerase,
- Deoxy nucleotide triphosphates (dNTPs)
- Reaction buffer.

Uses of Asymmetric PCR

 It finds use in some types of sequencing and hybridization probing where having only one of the two complementary strands is required.

Microbiology application:

- Identification of bacteria:
- Slow growing bacteria
- Bacterial antibiotic resistance genes.
- VIRUS: almost all viruses identification and viral load.
- FUNGUS
- Aspergillus spp
- Candida spp
- Plasmodium spp
- Trypanosoma spp

Forensic Application of PCR

- Forensic applications: DNA finger printing
- Detection of Mutations: Inherited disorders & carriers Track DNA abnormalities
- Prenatal diagnosis of genetic disorders.
- Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening
- Drug discovery
- Classification of organisms Genotyping

Nested PCR differ Multiplex PCR

NESTED PCR	MULTIPLEX PCR
Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR. Very low probability of nonspecific amplification.	Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences

Amplified fragment length polymorphism (AFLP) PCR

METHODOLOGY

- Genomic DNA is digested with one or more restriction enzymes. tetracutter (MseI) and a hexacutter (EcoRI).
- Ligation of linkers to all restriction fragments. Pre-selective PCR is performed using primers which match the linkers and restriction site specific sequences.
- Electrophoretic separation and amplicons on a gel matrix, followed by visualization of the band pattern.

AFLP PCR

- AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA.
- AFLP can be also used for genotyping individuals for a large number of loci.

Hot start PCR

- This is a technique that reduces nonspecific amplification during the initial set up stages of the PCR.

- The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C).
- Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature.
- This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step.
- DNA Polymerase- Eubacterial type I DNA polymerase, Pfu.
- These thermophilic DNA polymerases show a very small polymerase activity at room temperature.

Differentiate between the each Step of PCR

I. Denaturation	II. Annealing	III. Extension
• First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20- 45 seconds.	• The reaction temperature is lowered to 40-65°C for 20-40 seconds.	• The temperature at this step depends on the type of DNA polymerase used.

• It causes melting of DNA template yielding single strands of DNA.	• Annealing of the primers to the single stranded DNA template.	• Taq polymerase has its optimum activity temperature at 70- 80°C. • Commonly 72°C is used. • At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's.

Mini Primer PCR:

 Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.

 This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene.

 PCR that can extend from short primers

Conventional PCR (Real Time PCR or q-PCR) & Ingredients

REAL TIME PCR

 Real Time PCR is a technique in which fluorophores bind to specific target regions of amplicons to produce fluorescence during PCR.

 Real Time PCR is used to measure the quantity of a PCR product.

 The fluorescence, measured in Real Time, is detected in a PCR cycler with an inbuilt filter flurometer.

 It is the method of choice to quantitatively measure starting amounts of DNA, cDNA or RNA

Quantitative real-time PCR is often confusingly known as RT-PCR (Real Time PCR) or RQPCR.

• QRT-PCR or RTQ-PCR are more appropriate contractions.

Ingredients

• QRT-PCR methods use fluorescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.

Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample

What are the Limitations of PCR?

- Contamination risk
- Primer complexities
- Primer-binding site complexities
- Amplifies rare species
- Detection method

Methylation PCR (MSP)

 Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA.

 Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

 Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

 MSP used in quantitative PCR provides information about methylation state of a given CpG island.

Different between Colony PCR, In Situ PCR, Asymmetric PCR:

Colony PCR	In Situ PCR	Asymmetric PCR
The screening of bacterial or yeast clones for correct ligation or plasmid products.	In Situ PCR (ISH) is a polymerase chain reaction that actually takes	Asymmetric PCR is used to amplify one strand of the original DNA

	place inside the cell on a slide. □ In situ PCR amplification can be performed on fixed tissue or cells .	more than the other. □ It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.

Negative control and blank reaction

NEGATIVE CONTROL

□ Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

BLANK REACTION

□ Controls for contamination contains all reagents except DNA template.

Define types of (R-M)

Type I

♣ Type I systems were the first to be characterized from E. coli K 12

♣ The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit

Type II

♣ Most of the useful R- M system is Type II

♣ Type II enzymes recognize a defined sequence and cut within it

Type III

♣ Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value

Type IIs

♣ Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

Number and size of restriction fragments

The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut.

Four base recognition site occurs every 44 (256) bp

Six base recognition site occurs every 46 (4096) bp

Eight base recognition site occurs 48 (65,536) bp

Joining DNA molecule by alkaline phosphatase.

♣ An enzyme responsible for removing phosphate groups from many types of molecules including DNA

♣ Treatment of linearized plasmid vector DNA with alkaline phosphatase to remove 5'-terminal phosphate groups will prevent recircularization and plasmid dimer formation

Diff between DNA ligase and alkaline phosphatase.

DNA ligase	alkaline phosphatase
An enzyme that creates a phosphodiester bond between the 3' end of one	♣ An enzyme responsible for removing phosphate groups from many types of

DNA segment and the 5’ of another. DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands	molecules including DNA ♣ Treatment of linearized plasmid vector DNA with alkaline phosphatase to remove 5’-terminal phosphate groups will prevent recircularization and plasmid dimer formation

4 general characteristics of SNPs.

♣ IN DISEASE DIAGNOSIS
♣ IN FINDING PREDISPOSITION TO DISEASES
♣ IN DRUG DISCOVERY & DEVELOPMENT
♣ IN DRUG RESPONSES
♣ INVESTIGATION OF MIGRATION PATTERNS

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer

- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

Difference between southern blotting and northern blotting

SOUTHERN BLOTTING	NORTHERN BLOTTING
The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975). Identify DNA sequence (gene) of interest. Identified DNA may be a small piece of DNA or a mutation.	Northern blot is a method used for detection of a specific RNA sequence in RNA samples. This uses Agarose/formaldehyde gel.

Southern blotting detail

 The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

 Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975).

 Other blotting methods with similar principles, but using protein or RNA, have been named in reference to Edwin Southern's name.

 Identify DNA sequence (gene) of interest.

 Identified DNA may be a small piece of DNA or a mutation.

VNTR

• DNA profiling uses repetitive sequences that are highly variable, called variable number tandem repeats (VNTRs), particularly short tandem repeats (STRs). VNTR loci are very similar between closely related humans.

• The analysis of variable number of tandem repeats (VNTRs), to detect the degree of relatedness to another

sequence of oligonucleotides , making them ideal for DNA fingerprinting.

 A Variable Number Tandem Repeat (or VNTR) is a location in a genome where a short nucleotide sequence is organized as a tandem repeat.

 These can be found on many chromosomes, and often show variations in length between individuals.

There are two principal families of VNTRs:

¬ Microsatellites.

¬Minisatellites.

Microsatellites, also known as Simple Sequence Repea ts (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA.

• A minisatellites (also referred as VNTR) is a section o f DNA that consists of a short series of bases 10– 60 base pairs.

• Their analysis is useful in genetics and biology resear ch, forensics, and DNA fingerprinting.

4 methods of joining DNA fragments.

Mainly three methods are used for joining DNA in vitro

(i)Joining covalently annealed cohesive ends by DNA ligase

(ii)Joining blunt-ended fragments by DNA ligase from phage T4 infected E.coli

(iii The third utilizes the enzyme terminal deoxynucleotidyltransferase to synthesize homopolymeric3’ single stranded tails at the ends of fragments

Applications of Western blot

- Analysis Of IgG Fractions purified from human plasma.
- Diagnosis of HIV by ELISA, involves the western blotting technique.
- Western blotting technique is also used to Detect Some Forms Of Lyme Disease.
- Western blotting technique is used in Definitive Test For BSE, which is commonly know as Mad cow disease.

 Confirmatory Test For Hepatitis-B involves western blotting technique.

 Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.

 This technique is also employed in The Gene Expression Studie

 very delicate and time consuming process. A minute imbalance at any level of the procedure can skew the results of the entire process.

 Incorrect labeling of the protein can happen due to the reaction of secondary antibody.

 Cause erroneous in bands or no bands due to insufficient transfer.

 Well trained technicians are required for this technique.

 Primary antibody availability is crucial.

 It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

 Western blotting technique is simply a way to identify unknown proteins on a polyacrylamide gel.

 It is sometimes called as protein blotting or immunoblotting.

 It is a widely used analytical technique in the fields of molecular biology, immunogenetics, and other biochemistry disciplines.

 Western blotting technique is also used in the field of medical diagnostics. i.e., in the analysis of various kinds of diseases.

 Apart from the limitations of western blotting, it is more helpful now a days. Hence, The discovery of Western blotting technique has become boon in the field of science & technology

DNA modifying enzymes

- Nucleases
- DNA Polymerase
- Reverse transcriptases
- DNA ligases

Explain different blotting techniques

	Northern Blotting	Southern Blotting	Western Blotting
Type of Molecule Detected	Northern blotting detects a specific RNA sequence from an RNA sample.	Southern blotting detects a specific DNA sequence from a DNA sample.	Western blotting detects a specific protein from a protein sample.
Type of Gel	This uses Agarose/formaldehyde gel.	This uses an Agarose gel.	This uses Polyacrylamide gel.
Blotting Method	This is a capillary transfer.	This is a capillary transfer.	This is an electric transfer.
Probes Used	cDNA or RNA probes labeled radioactively or nonradioactively.	DNA probes are labeled radioactively or non-radioactively.	Primary antibodies are used as probes

Detection System	This is done using an autoradiograph, or detection of light or color change.	This is done using an autoradiograph, detection of light or color change.	This is done using the detection of light or color change.

Techniques to get known polymorphism

- Hybridization Techniques
  - Micro arrays
  - Real time PCR
- Enzyme based Techniques
  - Nucleotide extension
  - Cleavage
  - Ligation
  - Reaction product detection and display
- Direct Sequencing
- Microarray

- Cleavage / Ligation
- Electrophoretic mobility assays

Denature step of DNA joining

I. Denaturation II.
   - First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20- 45 seconds.
   - It causes melting of DNA template yielding single strands of DNA.

Note on restrictions Fragment length polymorphism (RFLP) / Procedure of RFLP

• RFLP analyzes the length of the strands of the DNA molecules with repeating base pair patterns.

- The basic technique for detecting RFLPs involves fragmenting a sample of DNA by a restriction enzyme, which can recognize and cut DNA wherever a specific short sequence occurs, in a process known as a restriction digest.

- The resulting DNA fragments are then separated by length through a process known as agarose gel

electrophoresis, and transferred to a membrane via the Southern blot procedure.

 Hybridization of the membrane to a labeled DNA probe then determines the length of the fragments which are complementary to the probe.

 An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

Electrophoresis of DNA-3

 Electrophoresis is a separations technique that is based on the mobility of ions in an electric field. Positively charged ions migrate towards a negative electrode and negatively-charged ions migrate toward a positive electrode.

Ions have different migration rates depending on their total charge, size, and shape, and can therefore be separated.

Define western blotting and its applications

 A technique used to detect the presence of a specific protein in a complex protein mixture.

 To determine the molecular weight of a protein.

 To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled.

Applications of western blotting

 Western blots have become one of the most common analytical tools for the detection of viral proteins.

 Characterization of monoclonal and polyclonal antibody preparations and in determining the specificity of the immune response to viral antigens.

Define transfer factors of southern blotting

 Upward capillary transfer

 Downward capillary transfer

 Simultaneous transfer to two membranes

 Electrophoretic transfer

 Vacuum transfer

Define the type of pcr in which RNA used as a template

 Northern blot is a method used for detection of a specific RNA sequence in RNA samples. This uses Agarose/formaldehyde gel. This is a capillary transfer. cDNA or RNA probes labeled radioactively or nonradioactive. This is done using an autoradiograph, or detection of light or color change.

Define Northern Blotting applications

 Study of gene expression in eukaryotic cells.

 To measure the amount & size of RNAs transcribed from eukaryotic genes.

 To estimate the abundance of RNAs.

 Study of gene expression in eukaryotic cells.

 To measure the amount & size of RNAs transcribed from eukaryotic genes.

 To estimate the abundance of RNAs.

What is alu PCR.

 A method for DNA fingerprinting that uses primers selected from segments repeated throughout a genome to produce a unique fingerprint of amplified product lengths.

 The use of primers from a commonly repeated segment is called Alu-PCR, and can help to amplify sequences adjacent (or between) these repeats.

 PCR used to produce a unique fingerprints of amplified product lengths.

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